Analysis of Contemporary Economic Issues

Guidance 2: Patterns of World Trade, Migration, and FDI

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Outline

Basic Concepts of Trade

Definitions

Export (X) "Out of". Goods/services sold to foreign countries.

Import (M) "Bring in". Goods/services bought from foreign countries.

Trade Balance The difference between exports and imports:

$$\text{Trade Balance} = \text{Exports} - \text{Imports}$$

- **Trade Surplus**: Exports > Imports. Ex. China
- **Trade Deficit**: Imports > Exports. Ex. USA
- **Bilateral Trade Balance**: Difference between exports and imports between *two specific countries*. Ex. USA bilateral trade balance with China has been a trade deficit of +200B$ every year since 2005.

Macroeconomics View: GDP

The Gross Domestic Product (Y) can be decomposed as:

GDP Identity

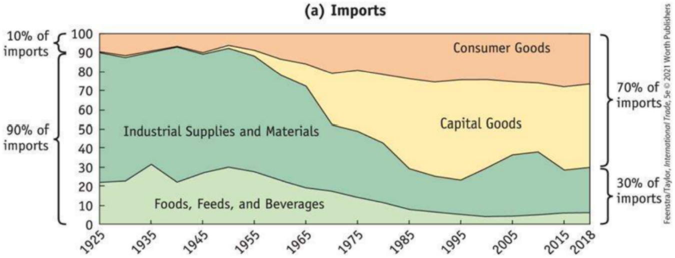
$$Y = C + I + G + EX - IM$$

- **C**: Consumers spending
- **I**: Firms investment
- **G**: Government spending
- **EX**: Exports
- **IM**: Imports

Fundamental Identity : (S-I)+(T-G)=EX - IM

- If (S-I) and (T-G) < 0, the country faces a trade deficit. There is not enough savings in the economy. This result highlights how the trade balance is determined by macroeconomic behavior.

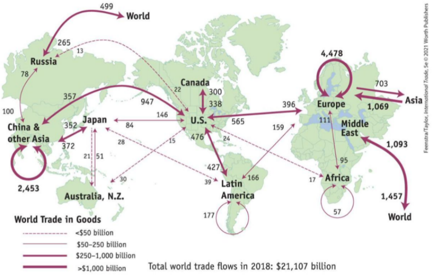
Evolution of Trade Structure (USA Case)



Structural Change (1925 vs 2018):

- **Before:** Dominance of *Industrial Supplies* and Materials.
- **After:** Shift towards *Capital Goods* and *Consumer Goods*.
- Foods, Feeds, and Beverages share has decreased significantly.
- **Note:** Service trading also increasing but not showed here.

Global Trade Flows



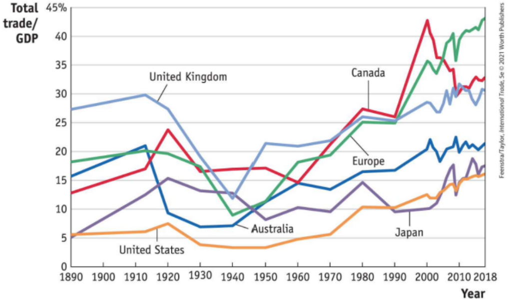
Key Observations:

- **Europe:** Largest amount of trade is *Intra-Europe* (Countries close to each other, easy to trade, EU market).
- **Americas:** Large trade flow within North/South America (US, Canada, Mexico - 2020 Agreement).

Global Insight:

- Trade occurs mostly between countries with **similar levels of industrialization**.
- Large trade between Europe and the US.
- Asian exports = 29% of world trade (2018).

A History of Globalization (Trade/GDP)



1. **1st Golden Age (1890s – WWI):** Improvements in transportation (steamship, railroad). $\rightarrow$ increase in trade
2. **Inter-war Period:** Sharp decline due to wars and Great Depression.
3. **Post-1950s:** Trade rebound ("Second Golden Age").
   Drivers $\rightarrow$ Reduced barriers (tariffs), containerization (1956), lower transport costs.

Note : Large countries tend to have small Trade/GDP ratios, and vice-versa.

Trade Barriers & Tariffs History

Historical Milestones

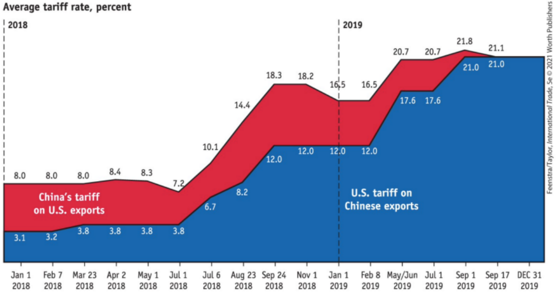
- **1865 (US Civil War):** High tariffs to raise revenue.
- **1890 (Tariff Act):** Raised tariffs to protect industries.
- **June 1930 (Smoot-Hawley Tariff):**
  - Raised tariffs up to 60% to protect farmers/industries.
  - **Consequence:** Trade war (Retaliation by Canada, France, etc.).

Modern Era: Since WWII, average world tariffs have drastically decreased (from $\approx 25\%$ to $< 5\%$), fostering globalization.

The US - China Trade War (2018-2019)

July 2018: Trump administration imposes tariffs against China.

- *Objective:* Force negotiations, lower Chinese tariffs on automobiles, increase US agricultural product exports.



Escalation:

- **US Tariffs on China:** Rose from $\approx 3.1\%$ (Jan 2018) to $\approx 21.0\%$ (Sep 2019).
- **China Tariffs on US:** Retaliation, rose from $\approx 8.0\%$ to $\approx 21.8\%$.

2025 Trump's tariffs

Timeline of US trade policy up until Sep. 2025 :

Date	Action / Event	Details
Feb 1, 2025	Tariffs announced on Canada & Mexico	25% tariffs on most goods from Canada/Mexico; 10% on energy. Linked to migration, border security, and trade imbalance.
Feb 13, 2025	"Reciprocal Trade and Tariffs" Memorandum	Directive to design 'reciprocal tariffs' responding to foreign trade barriers.
Mar 24, 2025	EO 14245: Venezuelan oil tariffs	25% tariff on goods from countries importing Venezuelan oil. Trade as foreign policy tool.
Apr 2, 2025	EO 14257: 'Liberation Day Tariffs'	Baseline 10% tariff on imports from nearly all countries; higher reciprocal rates (11–50%) for others.
Apr 5–9, 2025	Tariff implementation	10% baseline effective Apr 5; higher country-specific tariffs scheduled Apr 9 (some paused later).
May 12, 2025	Tariff adjustment with China	China agreed to reduce tariffs; U.S. moderated some planned rates.
Jul 31, 2025	EO 14326: Modification of reciprocal tariffs	Country-specific tariffs revived; framework deals offered to partners. Effective Aug 7.
Aug 6, 2025	Tariffs on India	25% tariff on all Indian goods, linked to purchases of Russian oil.
Sep 5, 2025	EO: Modifying Reciprocal Tariffs Scope	Modified Annex II of EO 14257; new procedures for trade/security agreements, adjusted coverage.

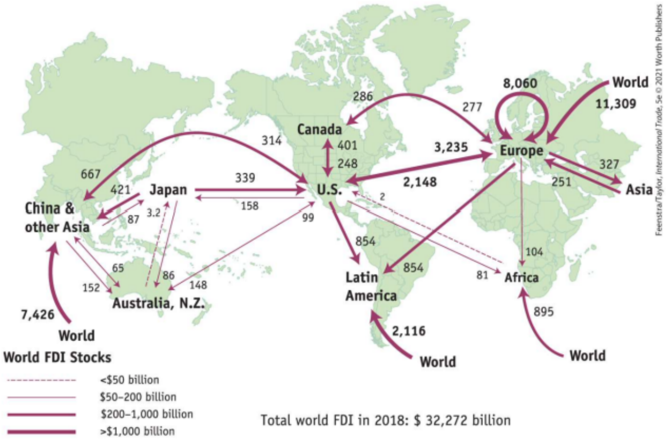
Since then :

- **New sets of worldwide tariffs :** Trump's administration collected around 300B$ in 2025.
- **Tariffs ruled out by Supreme Court :** Trump's use of emergency powers to enact the tariffs under IEEPA was not legal.
- **What about the revenue collected ?**

Migration vs. Trade

- **Direction:** Unilateral from poor to rich countries (whereas trade is bilateral).
- **Stats (2017):**
  - 53% of foreign-born lived in OECD.
  - 23% of OECD born were living another country
  - Most migration occurs *outside* OECD.
- **Theory:** Trade can act as a *substitute* for the movement of labor across borders by raising the living standards.

Foreign Direct Investment (FDI)



Total FDI (2018): $32.3 Trillion

Types of FDI

- **Horizontal FDI:** Between industrial countries (Rich to Rich). Dominant form.
  → mostly depends on trade frictions
- **Vertical FDI:** Industrial country owns plant in developing country.
  → mostly depends on production costs

Summary of Patterns

Globalization Components

Flow of goods/services, movement of people/firms, spread of culture/ideas, financial integration. → not a new trend ! started before WWI

- **Trade:** Mostly between industrial countries (Europe-US, Intra-Europe). Larger countries trade less relative to their GDP (internal market). Drivers = trade costs, TIC, transportation costs, preferences, size of economies.
- **Migration:** Mostly from developing to rich countries. Trade is a substitute for migration. Drivers = labor market capacity, culture, distance, restrictions, ...
- **FDI:** Unrestricted in developed countries. More FDI than trade in high-income countries; less in low-income. Drivers = business environment, production costs, ...

Gravity Equation's Intuition: Newton vs. Trade

Newton's Law of Gravity

$$F = G\frac{m_1 \times m_2}{d^2}$$

Objects with larger mass attract each other more; distance weakens attraction.

Variables:

- T_{ij}: Trade from country i to country j.
- Y_i: Production (GDP) of country i.
- E_j: Expenditure (GDP) of country j.
- D_{ij}: Distance between i and j.

Gravity Equation in Trade

Formula

$$T_{ij} = \frac{Y_i E_j}{D_{ij}^2}$$

Introduced by Jan Tinbergern :

- States that countries with larger GDPs, or that are close to each other, will have more trade between them.
- Important implications for trade models and trade flows.

Component 1: GDP

What is GDP?

- **Supply side:** Total value of final goods/services produced in a country (Y_i).
- **Demand side:** Total expenditure on all goods/services (E_j).

GDP measures the **economic size**.

- **Relationship:** GDP is *proportional* to trade. Why ?
- Large economies produce more (Supply/Exports) and buy more (Demand/Imports).
- *Data evidence:* Japan's trade with EU countries shows a clear positive correlation with their GDP (Slope ≈ 1).

Component 2: Distance

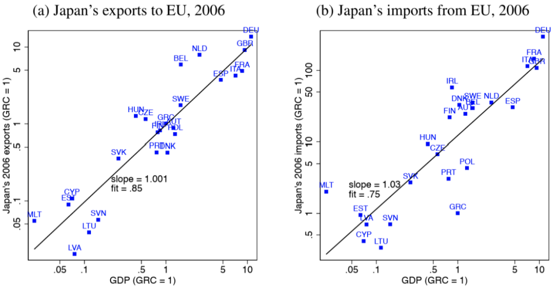
Trade is inversely proportional to distance.

What defines "Distance"?

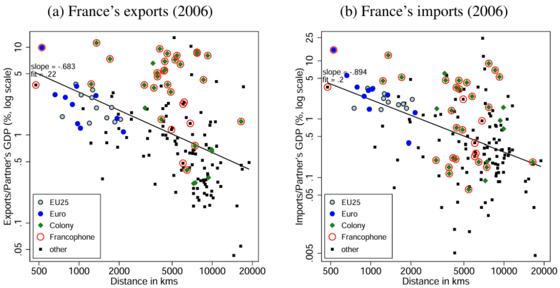
1. **Geographical/Physical:** Oceans, mountains, kilometers.
2. **Transportation Costs:** Infrastructure (ports, airports, rail), technology.
3. **Political Frictions:** Diplomatic ties, trade restrictions vs cooperation.
4. **Cultural Proximity:** Common language, colonial history, historic proximity.

Example: France trades more with neighbors and Francophone countries than distance alone would predict.

Evidence



Trade is proportional to size



Trade is inversely proportional to distance

Exercise: Applying the Gravity Equation

Data:

Country	Belgium (B)	Germany (G)	Mexico (M)
GDP	6	15	10

Distances: B-G: 5 | B-M: 100 | G-M: 100

Formula: $T_{ij} = \frac{Y_i E_j}{D_{ij}^2}$

Calculate Trade Flows

- Trade Belgium – Germany:
- Trade Germany – Mexico:
- Trade Mexico – Belgium:

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Calculate Trade Flows

- Trade Belgium – Germany: $= \frac{6 \times 15}{5^2} = \mathbf{3.6}$
- Trade Germany – Mexico:
- Trade Mexico – Belgium:

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Calculate Trade Flows

- Trade Belgium – Germany: $= \frac{6 \times 15}{5^2} = \mathbf{3.6}$
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- Trade Mexico – Belgium:

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Calculate Trade Flows

- Trade Belgium – Germany: $= \frac{6 \times 15}{5^2} = \mathbf{3.6}$
- Trade Germany – Mexico: $= \frac{15 \times 10}{10^2} = \mathbf{1.5}$
- Trade Mexico – Belgium: $= \frac{10 \times 6}{10^2} = \mathbf{0.6}$

Generalization

The generalized form uses parameters α, β, γ:

$$T_{ij} = K \frac{E_i^{\alpha} Y_j^{\beta}}{D_{ij}^{\gamma}}$$

Logarithmic Transformation:

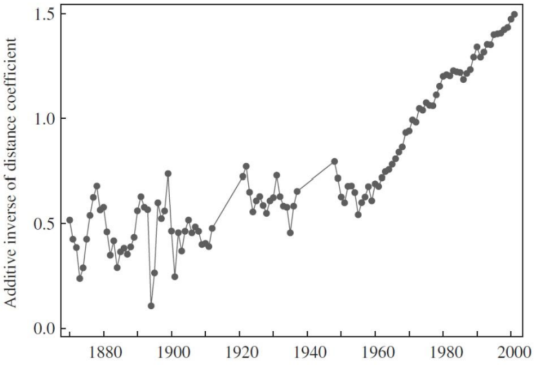
$$\ln T_{ij} = k + \alpha \ln E_i + \beta \ln Y_j - \gamma \ln D_{ij}$$

- More easier to manipulate with data.
- Coefficients can be interpreted as elasticities.
- γ is the elasticity of trade (w.r to distance).

Elasticity of trade

Elasticity of trade

Despite globalization and technology, the **elasticity of trade w.r to distance** γ (how much distance reduces trade) has **increased** over time. Distance matters *more* now than in 1880. Counterintuitive but true.



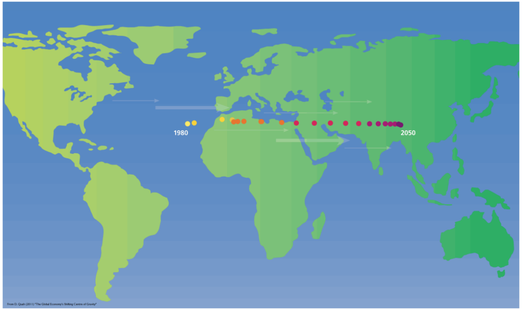
Why ?

- Change in the composition of trade (homogeneous vs differentiated goods).
- Regional trade blocs (EU, ASEAN, etc) encourage trade to be concentrated within those regions, making long distance less attractive.
- Trade heavily involves intermediate goods requiring reliable supply chains.

World's Economic Center of Gravity

The "weighted average" location of global economic activity is shifting.

- **Antiquity:** Mediterranean Sea.
- **XVI Century – 1980:** Atlantic Ocean (Europe/US dominance).
- **1980 – 2050:** Rapid shift towards **China** and Asia.



Modeling Migration with Gravity Equation

Migration Definitions

Movement away from usual residence.

- **Immigration** (In) vs **Emigration** (Out).
- Harder to model with Gravity: It's a discrete choice (0 or 1), involves personal preferences.
- Application started before trade.

Features of labor movements :

- **Positive selection** : the more educated are more likely to emigrate
- **Positive sorting** : the more educated go to countries with higher skill premia (= difference between wages of skilled and less skilled individuals)
- Wage differences sort migrants across destinations : workers with high education levels seek to maximize the returns on their human capital, finding higher rewards in nations with larger skill-related earning differences.

Modeling FDI with Gravity Equation

FDI Definitions

FDI = International movements of capital

- Investment into a country.
- Tangible investment and **NOT** portfolio investment.
- Takes the form of M&A mostly, i.e acquisition of another firm.

Since creation of the Euro :

- **Important growth driver** : increase in intra-EMU FDI stocks by 30%.
- **Increase** of FDI from EMU to non-EMU countries.

Modeling FDI with Gravity Equation

Table 6. *Euro effects outside the eurozone*

Dependent variable Period of estimation	ln(bilateral FDI outstock) 1992–2005	
Model	(1)	(2)
ln(parent country GDP)	1.337***	1.341***
	(0.203)	(0.198)
ln(host country GDP)	0.639***	0.640***
	(0.221)	(0.220)
EMU dummy	**0.344*****	**0.351*****
	(0.113)	(0.135)
EMU to non-EMU dummy	0.312***	
	(0.122)	
EMU to DNK-SWE-UK dummy		−0.025
		(0.160)
EMU to other-non-EMU dummy		0.322****
		(0.111)
Non-EMU to EMU dummy	−0.093	
	(0.092)	
DNK-SWE-UK to EMU dummy		0.161
		(0.127)
Other-non-EMU to EMU dummy		−0.143
		(0.102)
EU dummy	−0.011	−0.023
	(0.101)	(0.094)
Exchange-rate volatility	0.020	0.020
	(0.015)	(0.015)
ln(attractiveness of alternative locations)	−0.461***	−0.467***
	(0.108)	(0.108)
R^2-within	0.509	0.510
No. of observations	4,306	4,306
Country-pair dummies	Yes	Yes
Year dummies	Yes	Yes

Any Questions?

Clicker Questions

1. The majority of trade today is in:

- a. basic processed goods like steel.
- b. agricultural commodities.
- c. raw materials.
- d. highly processed consumer and capital goods.

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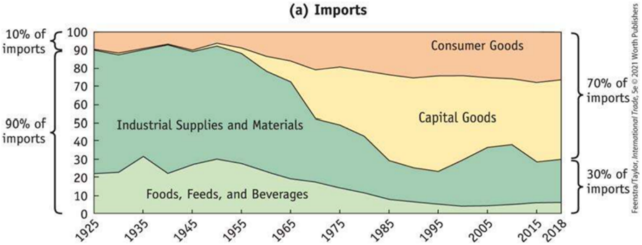

Clicker Questions

2. The largest amount of world trade (in U.S. dollars) is within:

- a. Asia.
- b. Europe
- c. The Americas
- d. Africa

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Shares of World Trade, Accounted for by Selected Regions, 2018

	Share of World Trade (%)		Share of World Trade (%)
Europe (internal trade)	21	Asia (exports)	29
Europe (internal) plus trade with the United States	26	Middle East and Russia (exports)	9
Americas (internal trade)	8	Africa (exports)	2
Europe and the Americas (exports)	49	Australia and New Zealand (exports)	2

Clicker Questions

3. A(n) .. is a tax on an imported good, and a(n) .. is a numerical limit on an imported good

- a. tariff; export quota
- b. import quota; export quota
- c. tariff; import quota
- d. import quota; tariff

Clicker Questions

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- a. tariff; export quota
- b. import quota; export quota
- **c. tariff; import quota**
- d. import quota; tariff

Export quota: numerical limit on an exported good.

Clicker Questions

4. Migrants typically choose .. countries to immigrate to.

- a. low-income
- b. middle-income
- c. high-income
- d. none of the above

Clicker Questions

4. Migrants typically choose .. countries to immigrate to.

- a. low-income
- b. middle-income
- c. **high-income**
- d. none of the above

Migration decisions mainly based on: labor market capacity, cultural aspects, distance, opportunities, ...

Clicker Questions

5. The majority of FDI occurs:

- a. between industrial countries.
- b. between high-income and low-income countries.
- c. between middle-income and low-income countries.
- d. between countries in Africa.

Clicker Questions

5. The majority of FDI occurs:

- a. **between industrial countries.**
- b. between high-income and low-income countries.
- c. between middle-income and low-income countries.
- d. between countries in Africa.

Horizontal FDI is the dominant form: 25% of the total world stock is located in European countries.

Clicker Questions

6. A firm's foreign direct investment decisions are, in the case of horizontal FDI, strongly influenced by .. and, in the case of vertical FDI, strongly influenced by .. .

- a. materials costs; labor costs
- b. trade costs; production costs
- c. labor costs; trade costs
- d. production costs; trade costs
- e. production costs; material costs